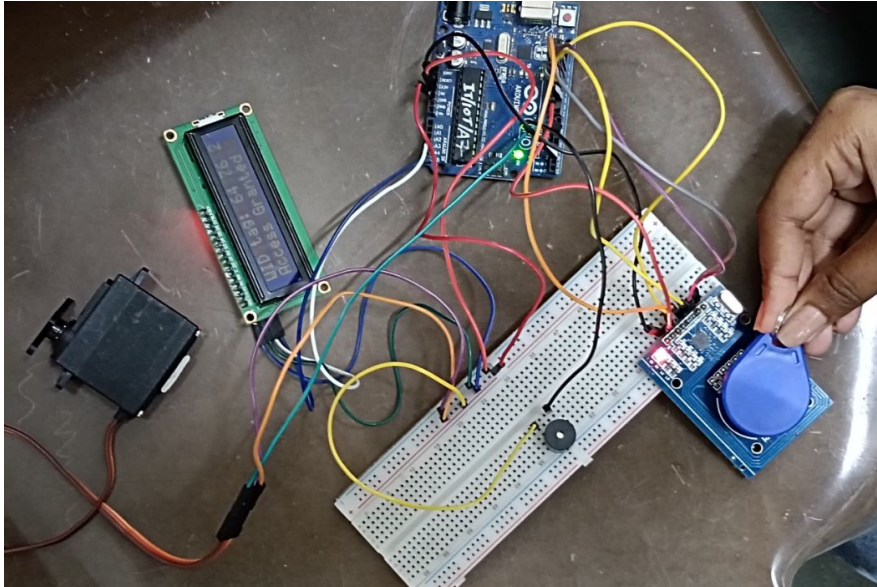


WIRELESS TECHNOLOGY

SMART TOLL TAX SYSTEM USING ARDUINO



Student Name : Ria Chaudhari
Class / Roll no: D15A / 07

Student Name: Eesha Chavan
Class / Roll no: D15A / 08

Student Name: Shraeyaa Dhaigude
Class / Roll no: D15A / 15

INTRODUCTION

- **Purpose:**
Automates toll collection by replacing manual methods with a digital, contactless system.
- **Technology Used:**
Arduino Uno, RFID module (RC522), Servo motor, Buzzer, LCD display.
- **Communication:**
RFID tag data is read and verified by the Arduino for access control.
- **Real-Time Action:**
Automatically opens the gate for authorized vehicles and logs the event.
- **Additional Feature:**
Buzzer alerts and LCD display provide visual and audio feedback on access status.
- **IoT Application:**
Integrates identification, automation, and monitoring for smart transportation.
- **Use Cases:**
Ideal for highways, residential complexes, college campuses, and parking systems to streamline vehicle access and reduce human intervention.

COMPONENTS REQUIRED

Component	Description
Arduino Uno	Microcontroller that controls the entire system
RFID Module (RC522)	Reads RFID tags for identifying authorized vehicles
RFID Tags	Unique ID tags assigned to vehicles
Servo Motor	Acts as the toll gate arm, opens/closes based on RFID validation
LCD Display (I2C)	Displays status messages like "Access Granted" or "Denied"
Buzzer	Provides sound feedback for successful or failed access
Jumper Wires	Used for connecting components
Breadboard	For prototyping and circuit assembly
Power Supply	9V Battery or USB power for running the system
LED Indicator (Optional)	Shows gate status visually (Green = Open, Red = Closed)

RFID: Radio Frequency Identification

RFID is a wireless technology used to identify and track tags attached to objects using radio waves. In this project, RFID cards act as vehicle identifiers. When an RFID tag comes in range of the RFID reader, it sends a unique ID to the Arduino, which then verifies access and triggers the toll gate mechanism (servo motor). This enables contactless, fast, and secure toll collection.

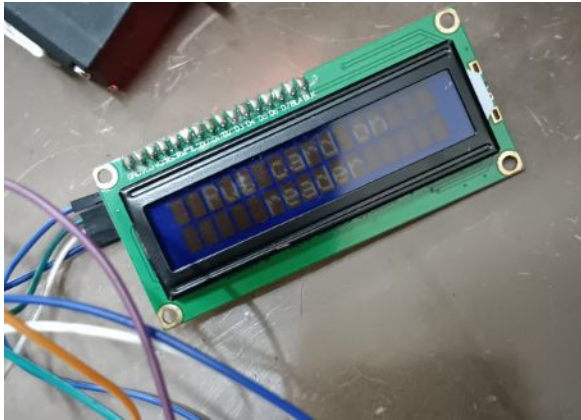


WORKING PRINCIPLE OF RFID

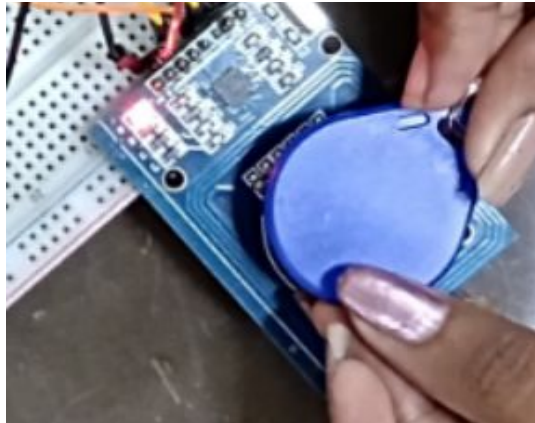
- **Tag and Reader Interaction:**
RFID systems consist of two main parts: the **RFID tag** (attached to a vehicle) and the **RFID reader** (installed at the toll booth).
- **Signal Transmission:**
The reader emits a radio frequency signal that powers the RFID tag when it enters the reader's range.
- **Data Exchange:**
The tag transmits its unique ID back to the reader via radio waves.
- **Authentication and Action:**
The Arduino processes the ID. If it matches an authorized ID, it activates the **servo motor** to open the toll gate.
- **Automation:**
The entire process is **contactless**, quick, and efficient—reducing congestion and manual errors.

BASIC WORKING

Message displays “Put Card on reader”



RFID tag is being read



Message displays “Access Granted” if ID is recognized or “Access Denied” otherwise

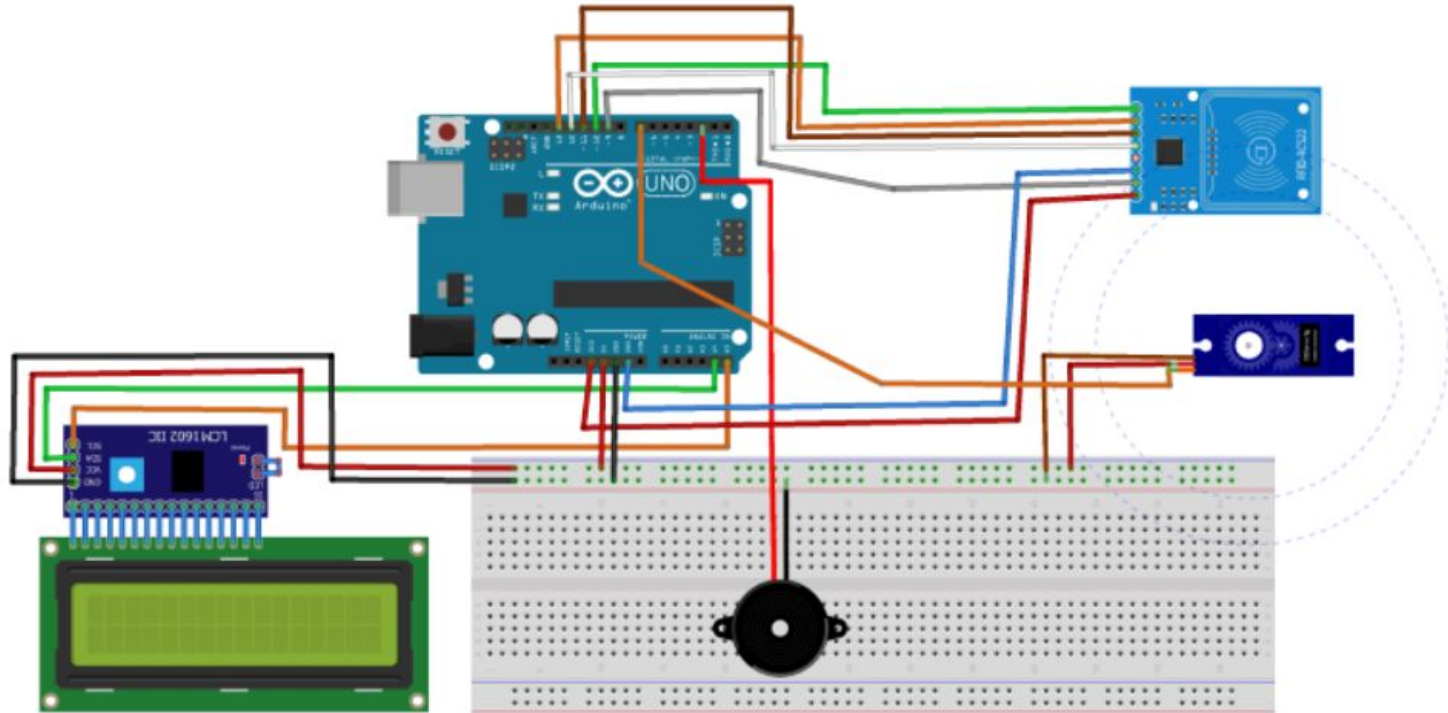


WORKING

Step Description

- An RFID card is placed near the RFID Reader (RC522) by the vehicle.
- The Arduino Uno reads the unique ID from the RFID tag.
- The ID is matched with pre-authorized entries stored in the Arduino code.
- If matched, the servo motor rotates to lift the toll gate (e.g., 90°).
- A buzzer beeps and the LCD displays "Access Granted" with vehicle info.
- After a short delay (2–3 seconds), the servo returns to the closed position.
- If the ID is not recognized, the LCD shows "Access Denied" and gate remains closed.

CIRCUIT DIAGRAM



PINS AND CONNECTIONS

RFID Reader to Arduino (SPI Protocol)

RFID Reader Pin	Arduino Pin	Description
SDA	10	Slave Select (SS)
SCK	13	Serial Clock (SCK)
MOSI	11	Master Out Slave In (MOSI)
MISO	12	Master In Slave Out (MISO)
GND	GND	Ground
3.3V	3.3V	Power Supply

LCD Display to Arduino (I2C Protocol)

LCD Display Pin	Arduino Pin	Description
SCL	A5	Serial Clock Line (SCL)
SDA	A4	Serial Data Line (SDA)
VCC	5V	Power Supply
GND	GND	Ground

PINS AND CONNECTIONS

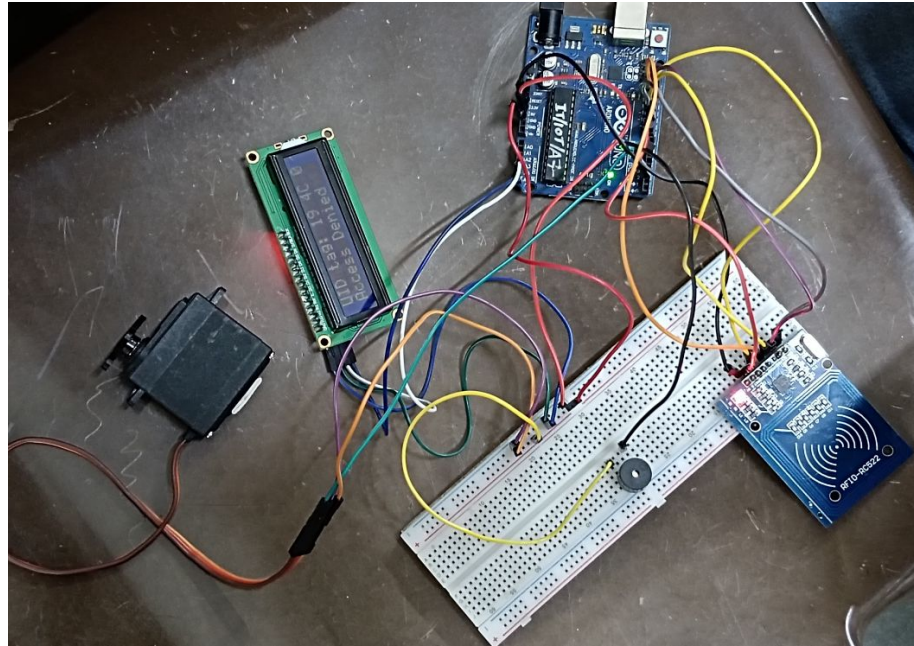
Servo Motor to Arduino

Servo Motor Pin	Arduino Pin	Description
Control Pin	D7	Signal Control
Power	5V	Power Supply
Ground	GND	Ground

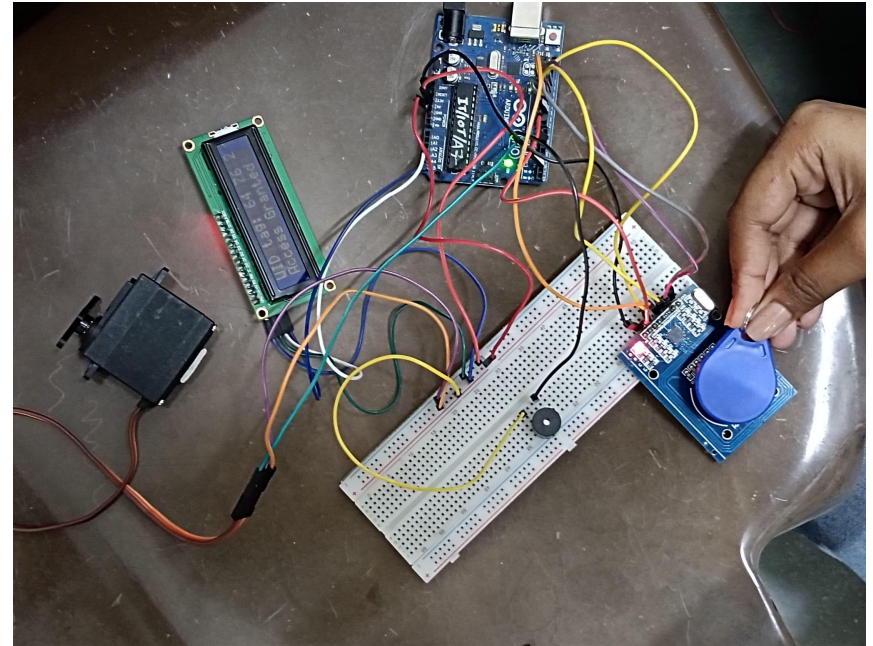
Buzzer to Arduino

Buzzer Pin	Arduino Pin	Description
+ (Positive)	D2	Signal Input
- (Negative)	GND	Ground

IMAGES OF IMPLEMENTATION



“Access Denied”



“Access Granted”

APPLICATIONS

- **Highways & Expressways** – For automatic toll collection and reducing traffic congestion.
- **Smart Cities** – Integrated into intelligent transport systems for smoother vehicular flow.
- **College/University Campuses** – To manage entry and exit of authorized vehicles efficiently.
- **Gated Communities & Apartments** – For secure and automated access to residents.
- **Corporate & Industrial Campuses** – To streamline vehicle access and maintain security logs.
- **Parking Lots** – For automated fee collection and gate control in commercial areas.

“Thank You”

[LINK TO REPORT](#)